

Methods in Java

Creating Methods

- Collection of statements grouped together to perform an operation

```
modifier returnType nameOfMethod (Parameter List) {  
  
    // method body  
  
}
```

- **modifier**: define the access type of the method
- **returnType**: Method may return a value
- **nameOfMethod**: this is the method name
- **Parameter List**: The list of parameters, it is the type, order, and number of parameters of a method
- **method body**: The method body defines what the method does with statements

Example

```
public static int minFunction(int n1, int n2) {  
    int min;  
    if (n1 > n2)  
        min = n2;  
    else  
        min = n1;  
    return min;  
}
```

Method Calling

There are two ways

returns nothing

```
System.out.println("This is tutorialspoint.com!");
```

returning a value

```
int result = sum(6, 9);
```

Example

```
public class ExampleMinNumber{  
    public static void main(String[] args) {  
        int a = 11;  
        int b = 6;  
        int c = minFunction(a, b);  
        System.out.println("Minimum Value = " + c);  
    }  
    /** returns the minimum of two numbers */  
}
```

```
/** returns the minimum of two numbers */  
public static int minFunction(int n1, int n2) {  
    int min;  
    if (n1 > n2)  
        min = n2;  
    else  
        min = n1;  
    return min; }  
}
```

Void Keyword

Allows to create methods which do not return a value

Function calling

```
public static void main(String[] args)
{ methodRankPoints (255.7); }
```

Function definition

```
public static void methodRankPoints (double points)
{...}
```


Passing Parameters by Value

Calling method with a parameter

```
public class SwappingExample {  
    public static void main(String[] args)  
    {  
        int a = 30; int b = 45;  
        swapFunction(a, b);  
    }  
    public static void swapFunction(int a, int b)  
    {...}  
}
```

Method Overloading

A class has two or more methods by same name but different parameters

```
public class ExampleOverloading {  
  
    public static int minFunction(int n1, int n2)  
    {  
    }  
  
    public static double minFunction(double n1, double n2)  
    {  
    }  
  
}
```